AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 5: Projectors; Conditioning and Condition Number

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Outline

1. **Projectors (NLA §6)**
2. Conditioning and Condition Numbers (NLA §12)

Projectors (Projection Matrices)

- A *projector* (also known as a *projection matrix*) $P \in \mathbb{R}^{m \times m}$ (or $P \in \mathbb{C}^{m \times m}$) satisfies $P^2 = P$. They are also said to be *idempotent*.
  - *Orthogonal* projector.
  - *Oblique* projector.
- An *orthogonal projector* is one that projects onto a subspace S_1 along a subspace S_2, where S_1 and S_2 are orthogonal.
  - $S_1 = \text{range}(P)$.
  - $S_2 = \text{null}(P)$.
- Example: Consider the matrix $P = \begin{bmatrix} 0 & 0 \\ \alpha & 1 \end{bmatrix}$.
  - P is an oblique projector if $\alpha \neq 0$.
  - P is an orthogonal projector if $\alpha = 0$.

Orthogonal Projector

Theorem

A projector $P \in \mathbb{R}^{m \times m}$ is orthogonal if and only if $P = P^{\mathsf{T}}$.

- **Remark**: An alternative definition of *orthogonal projection* is $P^2 = P$ and $P = P^{\mathsf{T}}$. It projects onto $S = \mathrm{range}(P)$.
- For complex matrices, we must replace $P = P^{\mathsf{T}}$ with $P = P^{\mathsf{H}}$.

Orthogonal Projector (Proof)

Proof

"If" direction: If $P = P^\top$, then

$$(Px)^\top (I - P)y = x^\top (P^\top - P^\top P)y = x^\top (P - P^2)y = 0.$$

"Only if" direction: Use the SVD. Suppose P projects onto S_1 along S_2 where $S_1 \perp S_2$, and S_1 has dimension n. Let $\{q_1, \dots, q_n\}$ be an orthonormal basis of S_1 and $\{q_{n+1}, \dots, q_m\}$ be a basis for S_2. Let Q be an orthogonal matrix whose jth column is q_j. We have $PQ = (q_1, q_2, \dots, q_n, 0, \dots, 0)$, so

$$Q^\top PQ = \mathrm{diag}(\underbrace{1, 1, \dots, 1}_{n}, 0, \dots) = \Sigma,$$

and $P = Q\Sigma Q^\top$.

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Discussion: Are orthogonal projectors orthogonal matrices?

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Complementary Projectors

- Complementary projectors: P versus $I - P$.
- What space does $I - P$ project onto?
  - Answer: $\text{null}(P)$.
  - $\text{range}(I - P) \supseteq \text{null}(P)$ because $Pv = 0 \Rightarrow (I - P)v = v$.
  - $\text{range}(I - P) \subseteq \text{null}(P)$ because for any v,

$$(I - P)v = v - Pv \in \text{null}(P).$$

- A projector separates $\mathbb{R}^m$ into two complementary subspaces: the range space and the null space (i.e., $\text{range}(P) + \text{null}(P) = \mathbb{R}^m$ and $\text{range}(P) \cap \text{null}(P) = \{0\}$ for projector $P \in \mathbb{R}^{m \times m}$).
- It projects onto the range space along the null space.
  - In other words, $x = Px + r$, where $r = (I - P)x \in \text{null}(P)$.
- **Discussion**: Are the range space and null space of a projector orthogonal to each other?

Uniqueness of Orthogonal Projectors

- The orthogonal projector for a subspace is unique.
- In other words, let $S \subseteq \mathbb{R}^m$ be a subspace. If P_1 and P_2 are both orthogonal projectors onto S, then $P_1 = P_2$.
- **Proof**: For any $z \in \mathbb{R}^m$,

$$
\begin{aligned}
\|(P_1 - P_2)z\|_2^2 &= z^\top (P_1 - P_2)^\top (P_1 - P_2)z \\
&= z^\top (P_1 - P_2)(P_1 - P_2)z \\
&= z^\top P_1 (P_1 - P_2)z - z^\top P_2 (P_1 - P_2)z \\
&= z^\top P_1 (I - P_2)z + z^\top P_2 (I - P_1)z \\
&= (P_1 z)^\top (I - P_2)z + (P_2 z)^\top (I - P_1)z \\
&= 0.
\end{aligned}
$$

Therefore, $\|P_1 - P_2\|_2 = 0$, implying $P_1 = P_2$.

Projections with Orthonormal Basis

- Given a unit vector q, we define $P_q = qq^\top$ and $P_{\perp q} = I - P_q$.
- Given any matrix $Q \in \mathbb{R}^{m\times n}$ whose columns q_j are orthonormal,

$$P = QQ^\top = \sum_j q_j q_j^\top$$

  is the orthogonal projector onto $\mathrm{range}(Q)$.
- SVD-related projections:
  - Suppose $A = U\Sigma V^\top \in \mathbb{R}^{m\times n}$ is the SVD of A, and let $r = \mathrm{rank}(A)$.
  - Partition U and V as

$$U = \begin{array}{c} \left[\begin{array}{c|c} U_r & \tilde{U}_r \end{array}\right] \\ \begin{array}{cc} r & m-r \end{array} \end{array}, \quad V = \begin{array}{c} \left[\begin{array}{c|c} V_r & \tilde{V}_r \end{array}\right] \\ \begin{array}{cc} r & n-r \end{array} \end{array}.$$

  - What do $U_r U_r^\top$, $\tilde{U}_r \tilde{U}_r^\top$, $V_r V_r^\top$, and $\tilde{V}_r \tilde{V}_r^\top$ project onto, respectively?

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  - What do $U_r U_r^\top$, $\tilde{U}_r \tilde{U}_r^\top$, $V_r V_r^\top$, and $\tilde{V}_r \tilde{V}_r^\top$ project onto, respectively?
  - Answer: $\mathrm{range}(A)$, $\mathrm{null}(A^\top)$, $\mathrm{range}(A^\top)$, and $\mathrm{null}(A)$.

Projection with Arbitrary Basis

- For an arbitrary vector $a \in \mathbb{R}^m$, we write $P_a = \frac{aa^\top}{a^\top a}$ and $P_{\perp a} = I - P_a$.
- Given any matrix $A \in \mathbb{R}^{m \times n}$ that **has full rank** with $m \geq n$, let $A = U\Sigma V^\top$ be its SVD. Then

$$P = UU^\top = A(A^\top A)^{-1}A^\top$$

  is the orthogonal projection onto $\mathrm{range}(A)$.
- The matrix $(A^\top A)^{-1}A^\top$ is called the *pseudo-inverse* of A, denoted as A^+.
- Therefore,

$$P = UU^\top = AA^+.$$

- In addition, $A^+A = I$.
- **Remark**: If $m < n$, $A^+ = A^\top(AA^\top)^{-1}$. Then $AA^+ = I$, and A^+A is the orthogonal projection onto $\mathrm{range}(A^\top)$.

Distance Between Subspaces

- Suppose S_1 and S_2 are subspaces of $\mathbb{R}^m$ with $\dim(S_1) = \dim(S_2)$, and P_i is the orthogonal projection onto S_i.
- The *distance* between S_1 and S_2 is defined as

$$\text{dist}(S_1, S_2) = \|P_1 - P_2\|_2.$$

- Suppose $W = [\underset{k}{W_1} \mid \underset{n-k}{W_2}]$ and $Z = [\underset{k}{Z_1} \mid \underset{n-k}{Z_2}]$ are $n \times n$ orthogonal matrices. If $S_1 = \text{range}(W_1)$ and $S_2 = \text{range}(Z_1)$, then

$$\text{dist}(S_1, S_2) = \|W_1^\top Z_2\|_2 = \|Z_1^\top W_2\|_2.$$

  (proof omitted here)
- In general, $0 \leq \text{dist}(S_1, S_2) \leq 1$.
- **Discussion**: If S_1 and S_2 are lines in 2D, what is $\text{dist}(S_1, S_2)$?

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Overview of Error Analysis

- Error analysis is an important subject of numerical analysis.
- Given a problem f and an algorithm $\tilde{f}$ with an input x, the *absolute error* is $\|\tilde{f}(x) - f(x)\|$ and the *relative error* is $\|\tilde{f}(x) - f(x)\| / \|f(x)\|$.
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- What are the possible sources of errors?
  - Round-off error (input, computation) — the main concern of NLA.
  - Truncation (approximation) error — the main concern for AMS 527.
- We would like the solution to be *accurate*, i.e., to have small errors.
- Error depends on the property (*conditioning*) of the problem and the property (*stability*) of the algorithm.

Example Using SVD Analysis

- Suppose $A \in \mathbb{R}^{n\times n}$ is nonsingular, and let $A = U\Sigma V^\top$ be its SVD.
- Then

$$Ax = U\Sigma V^\top x = \sum_{i=1}^{n} \sigma_i v_i^\top x u_i,$$

  and

$$x = A^{-1}b = \left(U\Sigma V^\top\right)^{-1} b = \sum_{i=1}^{n} \frac{u_i^\top b}{\sigma_i} v_i.$$

- Whether matrix multiplication and linear systems are sensitive to small changes in A or b depends on the distribution of singular values; a nearly zero σ_n can amplify errors in b along the direction of u_n.
- How do we formalize this analysis?

Absolute Condition Number

- The condition number is a measure of the *sensitivity* of a problem.
- The *absolute condition number* of a problem f at x is

$$\hat{\kappa} = \lim_{\varepsilon \to 0} \sup_{\|\delta x\| \le \varepsilon} \frac{\|\delta f\|}{\|\delta x\|},$$

  where $\delta f = f(x + \delta x) - f(x)$.
- Less formally, $\hat{\kappa} = \sup_{\delta x} \frac{\|\delta f\|}{\|\delta x\|}$ for infinitesimally small δx.
- If f is differentiable, then

$$\hat{\kappa} = \|J(x)\|,$$

  where J is the Jacobian of f at x, with $J_{ij} = \partial f_i / \partial x_j$, and the matrix norm is induced by vector norms on δf and δx.
- **Discussion**: What is the absolute condition number of $f(x) = \alpha x$?
- **Discussion**: Is the absolute condition number scale dependent?

Relative Condition Number

- The *relative condition number* of f at x is

$$\kappa = \lim_{\varepsilon \to 0} \sup_{\|\delta x\| \le \varepsilon} \frac{\|\delta f\| / \|f(x)\|}{\|\delta x\| / \|x\|}.$$

- Less formally, $\kappa = \sup_{\delta x} \frac{\|\delta f\| / \|f(x)\|}{\|\delta x\| / \|x\|}$ for infinitesimally small δx.
- Note: We can use different types of norms to get different condition numbers.
- If f is differentiable, then

$$\kappa = \frac{\|J(x)\|}{\|f(x)\| / \|x\|}.$$

- **Discussion**: What is the relative condition number of $f(x) = \alpha x$?
- **Discussion**: Is the relative condition number scale dependent?
- In numerical analysis, we generally use the relative condition number.
- A problem is *well-conditioned* if κ is small and is *ill-conditioned* if κ is large.

Examples

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    - Note: We are considering the condition number of the problem for a given x.
  - The relative condition number is

$$\kappa = \frac{\|J\|}{\|f(x)\|/\|x\|} = \frac{1/(2\sqrt{x})}{\sqrt{x}/x} = \frac{1}{2}.$$

- Example: Function $f(x) = x_1 - x_2$, where $x = (x_1, x_2)^{\mathsf{T}}$.

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- Example: Function $f(x) = x_1 - x_2$, where $x = (x_1, x_2)^{\mathsf{T}}$.
  - The absolute condition number of f at x in the ∞-norm is $\hat{\kappa} = \|J\|_\infty = \|(1, -1)\|_\infty = 2$.
  - The relative condition number is

$$\kappa = \frac{\|J\|_\infty}{\|f(x)\|_\infty/\|x\|_\infty} = \frac{2}{|x_1 - x_2|/\max\{|x_1|, |x_2|\}}.$$

  - κ is arbitrarily large (f is ill-conditioned) if $x_1 \approx x_2$ (hazard of cancellation error).
- Note: From now on, we will talk only about the relative condition number.

Condition Number of Matrix

- Consider $f(x) = Ax$, with $A \in \mathbb{R}^{m \times n}$.

$$\kappa = \frac{\|J\|}{\|f(x)\|/\|x\|} = \frac{\|A\|\|x\|}{\|Ax\|}.$$

- If A is square and nonsingular, since $\|x\|/\|Ax\| \leq \|A^{-1}\|$,

$$\kappa \leq \|A\|\|A^{-1}\|.$$

- Note that for $f(b) = A^{-1}b$, its condition number is also $\kappa \leq \|A\|\|A^{-1}\|$.
- We define the *condition number of matrix* A as

$$\kappa(A) = \|A\|\|A^{-1}\|.$$

- It is the upper bound of the condition number of $f(x) = Ax$ for any x.
- For any induced matrix norm, $\kappa(I) = 1$ and $\kappa(A) \geq 1$.
- Note the distinction between the condition number of a *problem* (the map $f(x)$) and the condition number of a *problem instance* (the evaluation of $f(x)$ for specific x).

Geometric Interpretation of Condition Number

- Another way to interpret $\kappa(A)$ is

$$\kappa(A) = \sup_{\delta x, x} \frac{\|\delta f\| / \|\delta x\|}{\|f(x)\| / \|x\|} = \frac{\sup_{\delta x} \|A\delta x\| / \|\delta x\|}{\inf_{x} \|Ax\| / \|x\|}.$$

- **Question**: For what x and δx is the equality achieved?

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- **Question**: For what x and δx is the equality achieved?
  - Answer: When x is in the direction of minimum magnification, and δx is in the direction of maximum magnification.
- Define the *maximum magnification* of A as

$$\text{maxmag}(A) = \max_{\|x\|=1} \|Ax\|$$

  and the *minimum magnification* of A as

$$\text{minmag}(A) = \min_{\|x\|=1} \|Ax\|.$$

- Then the condition number of the matrix is $\kappa(A) = \text{maxmag}(A)/\,\text{minmag}(A)$.
- For the 2-norm, $\kappa(A) = \sigma_1/\sigma_n$, the ratio of the largest and smallest singular values.

Example of Ill-Conditioned Matrix

Example

Let $A = \begin{bmatrix} 1000 & 999 \\ 999 & 998 \end{bmatrix}$. It is easy to verify that $A^{-1} = \begin{bmatrix} -998 & 999 \\ 999 & -1000 \end{bmatrix}$. So

$$\kappa_\infty(A) = \kappa_1(A) = 1999^2 = 3.996 \times 10^6.$$

Example of Ill-Conditioned Matrix

Example

A famous example of ill-conditioning is the Hilbert matrix, defined by $h_{ij} = 1/(i+j-1)$, $1 \le i, j \le n$. For example, for $n = 4$, we have

$$H_4 = \begin{bmatrix} 1 & 1/2 & 1/3 & 1/4 \\ 1/2 & 1/3 & 1/4 & 1/5 \\ 1/3 & 1/4 & 1/5 & 1/6 \\ 1/4 & 1/5 & 1/6 & 1/7 \end{bmatrix}.$$

This sequence of matrices is ill-conditioned even for quite small n. In particular, $\kappa_2(H_4) \approx 1.6 \times 10^4$, $\kappa_2(H_8) \approx 1.5 \times 10^{10}$, and $\kappa_2(H_{10}) \approx 1.6 \times 10^{13}$. The condition number grows exponentially in n.

Note that this example is an extreme case. Most matrices arising from practical applications are not nearly as bad.